

Goldsmiths, University of London

Institute of Management Studies

IM71065/IM53016: **Econometrics**

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Econometrics Project

In this Econometrics project you will learn:

- (i) How to **simulate** a data generating process (DGP) and how to create random variables in **R**.
- (ii) How to estimate parameters of a population equation using **OLS**.
- (iii) How to estimate parameters of a population equation using **2SLS** and **IVs**.
- (iv) How to estimate **robust HAC standard errors** and p-values for the estimators.
- (v) How to plot the data and your regression results in the same graph.
- (vi) How to compare the OLS and 2SLS estimates.
- (vii) How to deal with **endogeneity problems** using IVs and 2SLS.
- (viii) How to visualise the differences in the OLS and 2SLS estimators in the presence of regressor endogeneity.

Your task is to generate and simulate random variables for the population equation, which means that you will code the true **data generating process**. This will be a ‘simulated reality’ on your computer. The challenge is that you will introduce an **endogenous variable** in the data generating process. You will then estimate the parameters of this population equation using the ‘naïve’ **OLS estimator** and the correct **2SLS estimator** with instrumental variables (IVs). You will see that the OLS and the 2SLS estimators produce different estimates of the population equation, and that the 2SLS estimates are closer to the true parameter values. The purpose is for you to see in practice how an endogeneity problem makes the OLS estimator biased and inconsistent, and how the 2SLS estimator employs instrumental variables to get closer to the correct population value.

Now let us get to it. You must address all questions in your project.

Question 1 [20 points]

Simulate a data generating process with the following characteristics. These are the true population equations:

$$x = \gamma_0 + \gamma_1 z + \gamma_2 z^2 + \mu_1$$

$$\ln(y) = \beta_0 + \beta_1 \ln(x) + \mu_2$$

Where y is the **dependent variable**, x is an **endogenous** variable, and z is an **exogenous** variable (randomly assigned) with mean $\mu_z = 15$ and which affects y only through x . In your simulation in R, set $\beta_0 = 10$, $\beta_1 = 90$, $\gamma_0 = 100$, $\gamma_1 = 50$, and $\gamma_2 = -1$. Set the population size to $n = 2,000$.

The exogenous variable z enters twice in the first equation: firstly, in levels and, secondly, in quadratic form, implying that the relationship between x and z is non-linear. Is the relationship between z and x U-shaped or is it inverted U-shaped? Explain why the relationship between z and x has either a U shape or an inverted U shape.

The second equation is also non-linear, with natural logs on both sides. The non-linearities are not a problem because linear estimators such as OLS and 2SLS are linear in the parameters but not necessarily linear in the regressors. Thus, logs and quadratic regressors are not a problem in this case. But note that in R you will first simulate the values for y , and then generate $\ln(y)$.

Simulate the population errors μ_1 and μ_2 as random variables with a joint normal distribution, with mean zero and the variance-covariance matrix equal to sigma Σ , as follows:

$$(\mu_1, \mu_2) \sim N(0, \Sigma)$$

$$\Sigma = \begin{pmatrix} 1.0 & 0.5 \\ 0.5 & 1.0 \end{pmatrix}$$

Write the complete R code for this simulation using the correct parameter values. In your simulation, you will need to employ the function `set.seed()` using your own Goldsmiths ID number as the argument in the function. This will ensure that R generates random numbers that are unique to your Goldsmiths ID number.

Prepare a **joint plot** (2x2) of the following variables, using the **R code template on the VLE**:

Plot 1: scatterplot of x against z

Plot 2: scatterplot of x against $z + z^2$

Plot 3: scatterplot of y against x

Plot 4: scatterplot of $\ln(y)$ against $\ln(x)$

In questions 2 and 3 you will try to estimate the correct value of $\beta_1 = 90$ from the data generating process using the OLS and 2SLS estimators $\hat{\beta}_1$.

Question 2 [20 points]

Using the data that you simulated in question 1, you will now estimate β_1 using an estimator $\hat{\beta}_1$. In this question, estimate β_1 using the ‘naïve’ **OLS estimator** assuming (incorrectly) that regressor x is exogenous. Compute the standard errors and p-values for the estimator $\hat{\beta}_1$ using **robust HAC standard errors**.

Why is the OLS estimator ‘naïve’ in this case? Is the OLS estimator $\hat{\beta}_1$ unbiased and consistent? Why?

Plot and analyse the residuals from the OLS regression. Are the residuals well behaved?

Question 3 [20 points]

Estimate β_1 through the **2SLS estimator** $\hat{\beta}_1$ using z as the instrumental variable (IV). You will need to use the function **ivreg()** from package ‘AER’ in R. Note that you will use **two instruments** based on z , since z enters **twice** in the equation for x : first in levels and then squared. Compute the standard errors and p-values for the 2SLS estimator using **robust HAC standard errors**.

Is the 2SLS estimator $\hat{\beta}_1$ unbiased and consistent in this case? Why?

Plot and analyse the residuals from the 2SLS regression. Are the residuals well behaved?

Question 4 [20 points]

Create a joint single plot in R in which you include:

- (a) The data from the population equation: plot $\ln(y)$ in the vertical axis against $\ln(x)$ in the horizontal axis;
- (b) The true population function for y from question 1;
- (c) The OLS regression from question 2;
- (d) The 2SLS regression from question 3.

The graph must have the original (y, x) data plotted in the background in **natural logs**. Then add the OLS, 2SLS, and the population functions using **different colours** for each line. Add a legend with the corresponding colours of each estimator.

Explain **why** the 2SLS estimator is closer to the true population parameter β_1 than the naïve OLS estimator. Then show which estimator has the highest R^2 . Which estimator fits the data better? Is the estimator with the highest R^2 necessarily the correct one? **Why?**

Question 5 [20 points]

Replicate your project **10,000** times over, as in a Monte Carlo simulation. In each replication, compute the OLS and 2SLS estimators $\hat{\beta}_1$. **Plot the histogram of each estimator** across the 10,000 replications. Your histograms should show that:

- (i) The OLS estimator is a **biased** and **inconsistent** estimator of the true $\beta_1 = 90$;
- (ii) The 2SLS estimator is an **unbiased** and **consistent** estimator of the true $\beta_1 = 90$.

Then compute the average values and the standard deviations of each estimator. Show that the 2SLS estimator is **less efficient** (it has a **higher standard deviation**) than the OLS estimator.

In sum, you will use 10,000 Monte Carlo simulations of your project to **plot the frequency distributions** of the OLS and 2SLS estimators, showing that: (i) the OLS estimator is biased and inconsistent, but more efficient (lower standard deviation) than 2SLS; (ii) the 2SLS estimator is unbiased and consistent, but less efficient (higher standard deviation) than OLS.

Question 6 [challenge question for extra points]

Redo your entire project in **Python**. You can use ChatGPT to assist you in converting your R code to Python, but you must ensure that the Python code works and that it produces the correct results. Remember to add your **Goldsmiths student ID** as the seed number in the random number generator in Python as well. In **Jupyter Lab** it is even possible to run R and Python side by side in the same session, without changing the kernel. You could try that option in **Jupyter Lab** as well, since it will make things easier by keeping all code and outputs within a single file.

Final step

Submit your project as a **single PDF file** on the Moodle VLE page of the Econometrics module, using the appropriate Turnitin link. Your **full R code must be embedded** within the project. Do not add the R code as an attachment at the end of the project file.

As an **optional challenge**, you can code your entire project using either **Rmarkdown** or **Jupyter Lab** and render the project in PDF format. Or you can render it in HTML format, open the webpage file on your browser, and then save it as a PDF file.

On the Moodle VLE I will post a **template of the R code** that will help you to get started with the first question. The R code template will make things much easier for you.

You can use **ChatGPT** to help you write the R and Python code, but you must ensure that your code works and that it can create the plots for your project. I will check if your code really works when you submit your project on the VLE.

Use the **set.seed()** function using your own Goldsmiths ID number. The results in your project will be **unique to you and different from all other students**.

Good luck!